END-TO-END PROJECT: SUPERMARKET CUSTOMER RATING PREDICTION SYSTEM

Course: *Data Science and Machine Learning*

Project Type: *Predictive Analytics & Model Development*

Environment: *Kaggle / Jupyter Notebook*

Target Variable: *Rating (Regression/Classification)*

Serialization: *Joblib or Pickle*

Dataset: *https://www.kaggle.com/datasets/alexhuitron/supermarket-sales*

Time Frame: 2 Weeks

PROJECT OVERVIEW

The objective of this project is to build a machine learning pipeline to predict the ***customer satisfaction rating*** *of supermarket transactions. By analyzing factors such as product line, payment method, and transaction volume, the system aims to identify what drives a high-quality shopping experience.*

The student must:

- *Perform detailed exploratory data analysis (EDA) on retail transaction data.*
- *Handle categorical encoding for features like "Branch," "Customer type," and "Product line".*
- *Train and compare multiple machine learning models for regression (predicting the exact 1-10 score) or classification (predicting "High" vs "Low" rating).*
- *Evaluate models using appropriate performance metrics.*
- *Select and justify the best model for business insights.*
- *Serialize the trained model and preprocessing pipeline for future use.*

DATASET DESCRIPTION

- ***Dataset Name:*** *Supermarket Sales*
- ***Total Records:*** *1,000 transactions*
- ***Total Features:*** *17 columns*
- ***Target Variable: Rating*** *(Customer satisfaction rating).*

Key Features:

- ***Branch & City:*** *Location of the supercenter (Branches A, B, C).*
- ***Customer Type:*** *Member or Normal.*
- ***Gender:*** *Female or Male.*
- ***Product Line:*** *Category of product (e.g., Electronic accessories, Fashion accessories, Health and beauty).*
- ***Unit Price & Quantity:*** *Price per unit and number of items.*
- ***Tax 5%, Total, COGS:*** *Financial breakdown of the transaction.*

- ***Payment:*** *Method used (Cash, Ewallet, Credit card).*
- ***Date & Time:*** *When the transaction occurred.*

PHASE 1: DATA EXPLORATION AND PREPROCESSING

This phase must be implemented in a Jupyter Notebook (`supermarket_rating_analysis.ipynb`).

Step 1: Load and Inspect Data

- *Check for missing values*
- *Check for duplicate Invoice IDs.*
- *Display summary statistics (Mean rating is ~6.97).*

Step 2: Exploratory Data Analysis (EDA)

- ***Rating Distribution:*** *Plot a histogram of ratings to see the spread between 4 and 10.*
- ***Branch Comparison:*** *Average rating per branch to identify performance gaps.*
- ***Gender/Customer Type Analysis:*** *Compare satisfaction levels between Members and Normal customers.*
- ***Correlation Heatmap:*** *Identify relationships between "Total" sales, "Quantity," and the "Rating".*
- ***Time Series Analysis:*** *Check if ratings fluctuate based on the hour of the day or day of the week.*

Step 3: Feature Engineering & Preprocessing

- ***Categorical Encoding:*** *Apply One-Hot Encoding to "Branch," "Product line," and "Payment".*
- ***DateTime Features:*** *Extract "Hour" and "Weekday" from the Time and Date columns.*
- ***Feature Scaling:*** *Apply `StandardScaler` to numerical values like "Unit price" and "Total".*

Step 4: Train-Test Split

- *Split the data (e.g., 80/20) into training and testing sets.*

PHASE 2: MODEL TRAINING AND SELECTION

*The student must train at least three models to predict the **Rating**:*

1. ***Model 1: Linear Regression*** *(Baseline model for regression tasks).*
2. ***Model 2: Random Forest Regressor*** *(To capture non-linear relationships between product categories and ratings).*
3. ***Model 3: XGBoost or Gradient Boosting*** *(To optimize prediction accuracy).*

(Alternative: If treated as a classification task, models like Logistic Regression and Random Forest Classifier can be used by binning ratings into 'High' and 'Low').

MODEL EVALUATION METRICS

Because this is a regression problem (predicting a score), evaluation must include:

- ***Mean Absolute Error (MAE):*** *Average magnitude of errors.*
- ***Root Mean Squared Error (RMSE):*** *To penalize larger rating errors.*
- ***R-squared (R^2):*** *To determine how well the features explain the rating variance.*

The student must:

- *Present a comparison table of all models.*
- *Select the best model based on the lowest RMSE/MAE.*
- *Discuss which features (e.g., Payment method or Product line) were most influential in predicting high ratings.*

MODEL SERIALIZATION

- *Save the best-performing model using joblib or pickle.*
- *Save the preprocessing objects (encoders and scalers).*
- ***Files to be delivered:*** *rating_model.pkl and preprocessor.pkl.*

FINAL SUBMISSION REQUIREMENTS

1. ***Complete Jupyter Notebook:*** *Including EDA, data cleaning, and model training.*
2. ***Serialized Model Files:*** *The .pkl files for the model and scaler.*
3. ***Technical Report (1–3 pages):*** ** Insights from EDA (e.g., which branch has the highest satisfaction).*
    - *Model comparison table.*
    - *Final model justification.*
    - *Future improvements (e.g., collecting more sentiment-based data).*